AEROX – AIRCRAFT SOFTWARE CHANGE IMPACT ASSESSMENT

Case Study Report

Project Information	Details
Project Name	AeroX
Platform	Mendix Studio Pro 11.14.0 Beta
Application Type	Responsive Web Application
Domain	Aircraft Software and Data Systems
Deployment	Mendix Cloud

1. Introduction

Aircraft software systems are composed of multiple interconnected elements such as software components, systems, configurations, aircraft, and verification test cases. A change to one software component may therefore affect several related elements. Identifying these relationships manually can be time-consuming and may make impact information difficult to maintain in a structured and traceable form.

AeroX is a Mendix-based application developed to provide a structured workflow for aircraft software change impact assessment. The application allows users to maintain aircraft and software-related information, create software changes, automatically identify related impact items, and process reviewer decisions.

2. Problem Statement

When an aircraft software component is changed, engineers need to determine which systems, configurations, aircraft, and test cases may be affected. In a manually managed process, tracing these relationships can require checking multiple records and maintaining the results separately.

The problem addressed by AeroX is the lack of a simple, structured workflow for tracing the relationships of an aircraft software change and presenting the resulting impact information for review.

3. Objectives

- Develop a Mendix application for aircraft software change impact assessment.
- Create a structured domain model for aircraft, configurations, systems, software components, and test cases.
- Allow users to create and manage software changes.
- Automate impact tracing through Mendix microflows.
- Generate structured impact items for affected systems, configurations, aircraft, and test cases.
- Provide a reviewer workflow for approval or rejection.
- Provide dashboard views for software-change status and assessment impact levels.
- Deploy the completed application to Mendix Cloud.

4. Proposed Solution

AeroX provides a centralized web application in which aircraft-related software information is represented through entities and associations. When a user runs an impact assessment for a software change, the application follows the defined relationships to identify related engineering objects and creates corresponding impact items.

4.1 Application Workflow

1. Register aircraft and maintain aircraft status.
2. Create configurations and associate them with aircraft.
3. Create systems and associate them with configurations.
4. Register software components and associate them with systems.
5. Create test cases and associate them with systems.
6. Create a software change and select the related software component.
7. Run the Impact Assessment action.
8. Generate an ImpactAssessment and related ImpactItem records.
9. Review the assessment.
10. Approve or reject the assessment and update the related statuses.

5. Technology Used

Technology	Purpose
Mendix Studio Pro	Application development and modeling
Mendix Domain Model	Entity and relationship design
Mendix Microflows	Business logic and workflow automation
Responsive Web Pages	Application user interface
Atlas UI and SCSS	Interface styling and layout
Mendix Cloud	Application deployment
Git / GitHub	Version control and project repository

6. System Design

6.1 Domain Model

The AeroX domain model contains the main business entities required for software change traceability. The Mendix Account/User functionality is provided by the platform and is not represented as a custom AeroX entity.

Entity	Purpose
Aircraft	Stores aircraft identification, registration, model, and status.
Configuration	Stores aircraft configuration information and version details.
System	Represents aircraft systems such as flight control, navigation, and communication.
SoftwareComponent	Stores software component identification, version, description, and status.
TestCase	Stores verification test cases and their type/status.
SoftwareChange	Stores the proposed software change and its lifecycle

	status.
ImpactAssessment	Stores the generated assessment, impact level, status, and summary.
ImpactItem	Stores individual affected elements identified during assessment.
Review	Stores reviewer decision, comments, and review date.

6.2 Important Relationships

- One Aircraft can be associated with multiple Configurations.
- Configurations and Systems have a many-to-many relationship.
- Systems and Software Components have a many-to-many relationship.
- Systems and Test Cases have a many-to-many relationship.
- A SoftwareChange is associated with the SoftwareComponent being changed.
- A SoftwareChange can have related ImpactAssessment records.
- An ImpactAssessment contains related ImpactItem records.
- An ImpactAssessment can have related Review records.
- ImpactItems can reference Aircraft, Configuration, System, or TestCase objects.

7. Implementation

7.1 Management Functions

- Aircraft management
- Configuration management
- System management
- Software component management
- Test case management
- Software change management
- Impact assessment viewing
- Impact item viewing
- Review management

7.2 Software Change Creation

The Software Change page allows the user to enter the change ID, change title, current version, proposed version, change description, reason, status, creation date, and associated software component. A Run Impact Assessment button starts the automated assessment process.

7.3 Automated Impact Assessment

The MyFirstLogic microflow implements the automated assessment. It receives the current SoftwareChange object and creates an ImpactAssessment. It then retrieves the associated SoftwareComponent and traverses the related System, Configuration, Aircraft, and TestCase relationships.

11. Create and commit an ImpactAssessment.
12. Associate the assessment with the SoftwareChange.

13. Retrieve the related SoftwareComponent.
14. Retrieve related Systems.
15. Create System impact items.
16. Retrieve related Configurations and create Configuration impact items.
17. Retrieve the Aircraft associated with each Configuration and create Aircraft impact items.
18. Retrieve related Test Cases and create Test Case impact items.
19. Set the SoftwareChange status to Assessed.

7.4 Review and Approval

The ProcessReview microflow handles the reviewer decision. If the Review decision is Approved, the related ImpactAssessment and SoftwareChange are updated to Approved. If the decision is Rejected, both are updated to Rejected. This separates automated impact identification from the final human review decision.

8. User Interface and Dashboard

The application uses a responsive Mendix web interface with a dedicated AeroX home page and navigation to the management and workflow pages. The home page provides direct actions for creating a software change, viewing software changes, and viewing assessments.

The dashboard provides an operational view of the application through two line charts. The first presents Software Change Status categories such as Draft, Assessed, Under Review, Approved, and Rejected. The second presents Impact Assessment Impact Level categories such as Low, Moderate, and High. A Software Change data grid provides detailed operational records.

9. Testing and Validation

The completed application was validated using representative data and workflow scenarios.

Test Case	Expected Result	Result
Approved workflow – CHG001	SoftwareChange and ImpactAssessment become Approved after an approved review.	Verified
Rejected workflow – CHG002	SoftwareChange and ImpactAssessment become Rejected after a rejected review.	Verified
Impact tracing	System, Configuration, Aircraft, and Test Case impact items are generated.	Verified
Dashboard	Software-change and impact-level charts display populated data.	Verified
Cloud deployment	The published application opens successfully through the Mendix Cloud URL.	Verified

9.1 Representative Test Data

- Aircraft: AC001, Registration VT-AX01, Model A320.
- Configuration: CFG001, A320 Standard Configuration, Version 1.0.

- System: SYS001, Flight Control System.
- Software Component: Flight Control Software, Current Version 2.1.0.
- Test Case: TC001, Flight Control Functional Test.
- CHG001: Upgrade Flight Control Software – approved workflow test.
- CHG002: Upgrade Flight Control Software - Rejection Test – rejected workflow test.

10. Results

The completed AeroX application provides an end-to-end workflow for aircraft software change impact assessment. The application successfully stores the required engineering relationships, creates software changes, generates impact assessments and impact items through a microflow, and processes reviewer decisions.

The dashboard provides a concise operational view of software-change statuses and impact levels. The application was also successfully published to Mendix Cloud.

11. Benefits

- Provides structured traceability between software changes and related engineering elements.
- Reduces the need for manual relationship tracing.
- Generates consistent impact-assessment records.
- Separates automated assessment from reviewer decision-making.
- Provides a centralized record of changes, assessments, impact items, and reviews.
- Provides an operational dashboard for quick status visibility.

12. Limitations

- Impact assessment is based on predefined relationships in the AeroX domain model.
- The current application does not integrate with external aircraft engineering databases or live enterprise APIs.
- The current deployment is single-tenant.
- The dashboard provides operational analytics rather than predictive analytics.
- Advanced risk scoring and configurable assessment rules are not included in the current version.

13. Future Enhancements

- Introduce configurable impact-assessment rules and more detailed impact scoring.
- Add richer role-based access and permissions.
- Add assessment history and audit-log functionality.
- Integrate external engineering and configuration-management systems.
- Add advanced analytics by aircraft, system, software component, and change type.
- Add notifications for pending reviews.
- Extend the application for multi-tenant enterprise deployment if required.

14. Conclusion

AeroX demonstrates the use of Mendix to develop a structured aircraft software change impact assessment application. The solution combines domain modeling, associations, microflows, responsive pages, review workflows, and dashboards to support the process from software-change creation through impact identification and reviewer decision.

The completed prototype provides a practical foundation for managing software-change traceability in an aircraft software context and can be extended with advanced rules, analytics, integrations, and enterprise-level capabilities.

15. References

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